Lecture 4 - Measuring Sustainability - Assessment, KPIs and Life Cycle Thinking

Lecturer notes, slide by slide

With activities, student questions and model answers

Deck	ASM_Lecture 4_Measuring Sustainability Assessment KPIs and LCA_revised.pptx
Slides	28 - numbering matches the revised PowerPoint
Lecturer	Jenan Hussein
Revision	5 October 2026; content and visual approach adapted from approved Lecture 1
Teaching use	Read the summary first, then keep the slide cards beside you. Say gives the explanation; Ask and Tip support delivery.

What changed

Seven focused objectives; clearer links to business decisions; the original six-part sequence, cases and activities; corrected qualifications where benefits or rules were too absolute. Full teaching notes are also embedded in the PowerPoint.

Seminar connection

Seminar 3 (Product life cycle and its environmental impacts, Ecodesign). The second half of this lecture is the direct preparation for it. The dashboard workshop feeds the indicators section of the semester paper.

1. The lecture at a glance

The big idea

Measurement should help someone make a decision. The first half turns environmental, economic and social goals into indicators with a baseline, unit, source, update frequency and accountable owner. The second half uses life-cycle thinking to compare products, roofs and company facilities on an equal basis. Keep environmental LCA, financial appraisal and wider public benefits distinct, then explain how they inform the same decision.

Learning outcomes

1. Explain how measurement supports management decisions.
2. Select environmental, economic and social KPIs for GI.
3. Design a dashboard with units, data sources and owners.
4. Explain the four phases of Life Cycle Assessment.
5. Use a functional unit to compare alternatives fairly.
6. Apply ecodesign to products and business facilities.
7. Assess roof options using life-cycle impacts and wider benefits.

Words to explain early

Term	Plain-language meaning
KPI	A key performance indicator used to track a defined objective.
Baseline	The starting value and conditions used to assess change.
LCA	Life Cycle Assessment: a structured assessment of environmental impacts over a defined product or service life cycle.
Functional unit	The quantified service that makes alternatives comparable, such as waterproofing one square metre of roof for 50 years.
System boundary	The processes and life stages included in an assessment.
LCI and LCIA	Life-cycle inventory records inputs and outputs; impact assessment translates them into environmental impact categories.
Ecodesign	Design that considers environmental impacts across the life cycle.
Embodied and operational impacts	Impacts from materials and construction, and impacts during operation; define boundaries to avoid double counting.
GIS, IoT and digital twin	Spatial mapping, connected field sensors, and a model for examining system behaviour or scenarios.

Timing plan

These are the original suggested block allocations, retained for planning. Adjust them to your actual teaching slot. Individual slide timings are approximate; do not add them to the block totals.

Block	Slides	Minutes
Welcome, objectives, agenda	1-3	8
Part 1 - Why and how we measure	4-5	12
Part 2 - KPIs for Green Infrastructure (incl. Letná exercise)	6-8	25
Part 3 - Data and tools	9-10	12
Workshop: build a GI dashboard	11	30
Break	12	15-20
Part 4 - Life Cycle Assessment	13-16	30
Part 5 - Ecodesign	17-18	12
Part 6 - Lifecycle thinking for buildings and sites + green roof case	19-23	25
Activity: map a life cycle	24	15
Key messages, reading, questions	25-27	10

Before class

Open itreetools.org once so you can describe what the tool looks like (slide 10). Bring flip-chart paper or ask students to bring laptops for the dashboard workshop (slide 11). Think of one product you used this morning for the mug vs paper cup discussion (slide 16). Compare equal numbers and volumes of drinks, and discuss manufacturing, washing and reuse assumptions; there is no universal break-even number.

Use the revised deck named on this cover. Confirm the timetable, seminar deadline and assessment details with the course organiser. Distinguish classroom assumptions from observed results.

2. Slide-by-slide notes

Slide numbers and titles match the revised deck. These are suggested words and prompts, not a script to read verbatim.

01 Applied Sustainability in Business and Management (1 min)

Say: Welcome. Say that last time we talked about strategy and targets. A target without a number is a wish - today is about the numbers.

02 Learning Objectives (3 min)

Say: Read the seven objectives. Connect each to a decision a manager needs to make. The first half introduces the tools and business problem; the second half applies them to the cases and activities. Students will use the same tools for their Prague semester site.

03 Today's Lecture Agenda (2 min)

Say: Six parts; a longer workshop before the break and a short activity at the end.

04 Part 1 - Why and How We Measure (1 min)

Say: Announce the part.

05 You Manage What You Measure (10 min)

Say: Three blocks. Why indicators matter: targets without indicators are wishes, and investors, regulators and citizens want evidence. The measurement cycle: baseline, target, monitor, report, improve - draw it as a circle on the board. Good KPIs are relevant, measurable, comparable over time, and hard to game. Give an example of a bad KPI: 'number of trees planted' is easy to game (plant cheap saplings that die); 'tree canopy cover in five years' is better.

Ask: Give me one KPI that is easy to game. How would you fix it?

06 Part 2 - KPIs for Green Infrastructure (1 min)

Say: Announce the part.

07 Three Families of GI Indicators (10 min)

Say: Compare environmental, economic and social indicators. Every quantitative KPI needs a unit, boundary, baseline and collection method. Carbon sequestration is a flow per year; carbon stock is the amount stored at a point in time. For green-space access, define the distance and route method explicitly. For a business project, distinguish savings the company captures from public benefits.

08 Measuring a Park: Prague Example (10 min)

Say: Use Letná as the public case, then ask how a nearby café, hotel or employer could use the same evidence. Name one environmental, one economic and one social indicator, its data source, and the decision it supports. Treat an investment amount as a hypothetical classroom example, not a confirmed city budget.

09 Part 3 - Data and Tools (1 min)

Say: Announce the part.

10 The GI Measurement Toolkit (10 min)

Say: GIS can map green cover, heat and access, while sensors collect field measurements. i-Tree estimates tree benefits using data and models. Dashboards communicate results; digital twins test scenarios. The screenshot in this slide is only an example of a GIS interface: its map is Houston income, not a sustainability analysis of Prague.

11 Workshop: Build a Green Infrastructure Dashboard (30 min)

Say: Keep the 30-minute group workshop and the nine KPIs. For each, add a baseline and accountable owner to the original unit, source and frequency. The dashboard should help a site manager, mayor or investor decide what to do. Ask groups to separate measured values from estimates or missing data.

12 Break (20 min)

Say: After the break: from measuring a place to measuring a product - Life Cycle Assessment.

13 Part 4 - Life Cycle Assessment (1 min)

Say: Announce the part. Say the short form: LCA.

14 What is LCA? (8 min)

Say: Explain goal and scope, inventory, impact assessment and interpretation. LCA includes a defined function and boundary across life stages. Do not teach a fixed dominant stage for every car, shirt or building: the result depends on the technology, use, lifetime and impact category.

15 The Four Phases of LCA (8 min)

Say: Four phases, in order. Goal and scope: what are we assessing, for which decision, and where are the boundaries. Inventory: collect all inputs and outputs - energy, materials, emissions, waste. Impact assessment: translate the inventory into impacts such as climate, water, toxicity, land use. Interpretation: draw conclusions, test how sensitive the result is, recommend. Students should be able to name the four phases in order.

16 The Functional Unit (8 min)

Say: Compare equal service, such as keeping one square metre of roof waterproof for 50 years. For cups, specify a number and volume of drinks plus washing and reuse conditions. There is no universal break-even number for a ceramic mug; it depends on manufacturing, energy, washing and reuse.

Ask: What would you need to measure to make the comparison fair?

17 Part 5 - Ecodesign (1 min)

Say: Announce the part.

18 Ecodesign Principles (10 min)

Say: Design affects materials, repair, energy use and recovery. Avoid treating a single percentage as universal. Connect ecodesign to the R-ladder: repair and reuse require durable components, spare parts and a workable service model. Ask which changes a procurement or product manager can specify.

19 Part 6 - Lifecycle Thinking for Buildings and Sites (1 min)

Say: Announce the part: LCA is not only for products.

20 Life-Cycle Thinking for Business Facilities (5 min)

Say: A company can use the same life-cycle approach for a roof, car park or production site. Include materials, transport, irrigation, maintenance, replacement and disposal as relevant. Vegetation may store carbon or reduce cooling needs, but the net result must be calculated. Conventional infrastructure also provides essential services.

Source references: see the source list at the end of this guide.

21 Case Study: the Roof Above Your Head (2 min)

Say: Chicago City Hall again - a working green roof on a hundred-year-old building. Now we look at it with LCA eyes.

22 Green Roof and Conventional Roof (6 min)

Say: Compare the same waterproofing service and study period. Avoid fixed lifetimes as universal facts: actual life depends on the system and maintenance. Green roofs can add benefits but also weight, materials and upkeep. Check the building’s structural suitability before comparing costs.

Source references: see the source list at the end of this guide.

23 Assessing the Green Roof Business Case (6 min)

Say: Energy savings or a longer membrane life may compensate for added burdens, but the conclusion depends on the site and study. Keep financial appraisal, environmental LCA and wider social or ecological benefits distinct. Use sensitivity checks for uncertain lifetimes and energy savings.

Ask: Would the preferred option change if maintenance costs doubled or the energy mix changed?

Source references: see the source list at the end of this guide.

24 Classroom Activity: Map a Life Cycle (15 min)

Say: Pairs choose a product they used today, sketch its life cycle, mark the stage with the biggest impact, and propose one ecodesign change. Details in section 3.

25 Key Messages (4 min)

Say: Work through the eight messages. Ask students to connect each to a decision, example or tool from this lecture. Distinguish potential benefits from measured results.

26 Recommended Reading (2 min)

Say: Point to the Seminar 3 preview: students will apply life cycle mapping and ecodesign to their Prague site. Next lecture: stakeholders, communication and smart cities.

27 Any questions? (5 min)

Say: Take questions.

28 Image credits

Say: No need to show.

3. How to run the class activities

Workshop - Build a Green Infrastructure Dashboard (slide 11, 30 minutes)

1. Project groups work on their own semester site. Each group needs a large sheet of paper or one laptop.

2. 5 minutes: choose 3 environmental, 3 economic and 3 social KPIs from slide 7 or their own.

3. 10 minutes: for each KPI write the unit, the data source (sensor, survey, city data, i- Tree, property portal) and how often it updates (live, monthly, yearly).

4. 5 minutes: sketch the screen. Rule: the mayor must see the three most important numbers in the top row; the investor's numbers can be a second row.

5. 10 minutes: two-minute presentations. Ask each group one question: which of your nine numbers would change a decision, and whose decision?

Watch for: Groups pick KPIs they cannot get data for ('biodiversity index'). Ask: who measures this, how often, and what does it cost? Push them towards data that exists (visitor counts, temperature, property prices).

Map a Life Cycle (slide 24, 15 minutes)

1. Pairs choose one product they used today (phone, coffee, jeans, bus ticket).

2. 8 minutes: draw five boxes - raw materials, production, transport, use, end of life. Under each box write the main inputs (energy, materials, water) and outputs (emissions, waste).

3. Mark the box they think dominates the impact and write why.

4. Propose one ecodesign change from slide 18 and predict what it does to the dominant stage.

5. 5 minutes: three pairs present. Ask the room whether they agree with the dominant stage - this is often where students learn the most.

Watch for: Students guess that transport dominates. For most products it does not - materials or use phase do. Use this to show why LCA is needed instead of intuition.

Business decision prompt

For each proposal, ask students to name the payer, operating owner, affected stakeholders and one measurable outcome. Use assumptions openly when data is unavailable.

4. If a student asks

Do we have to do a real LCA in the seminar paper?

No. You should apply lifecycle thinking: describe the stages, say where the biggest impacts are and why, and propose changes. A full LCA needs software and data; the logic is what counts.

Is a green roof always better than a normal roof?

There is no universal winner. Compare equal service over the study period, including climate, structure, materials, maintenance, replacement and energy. Include runoff, cooling and habitat as appropriate, and state which benefits are outside the LCA.

Why not just measure everything?

Because measurement costs money and attention. A dashboard with 40 numbers is ignored. Choose the few KPIs that answer a real decision.

Where can I get data for my Prague site?

The city's open data portal (opendata.praha.cz) and IPR Prague maps for green cover and land use; property portals for prices; the Czech Hydrometeorological Institute for temperature; i-Tree for tree values; your own counts and short surveys for social data.

5. Questions with model answers

Use these for discussion, homework or test preparation. Model answers are short examples; other well-supported answers can also be valid.

1. Write the five steps of the measurement cycle in order and explain what a baseline is for.

Recall

Model answer: Baseline > Target > Monitor > Report > Improve. The baseline is the starting value; without it you cannot show whether anything improved.

2. Name the four qualities of a good KPI. Give one KPI that is easy to game and a better alternative.

Understand

Model answer: Relevant, measurable, comparable over time, hard to game. Example: 'trees planted' can be gamed with cheap saplings that die; 'canopy cover after five years' is harder to game.

3. Give three environmental, three economic and three social KPIs for a new park, each with a unit.

Recall

Model answer: Environmental: annual carbon sequestration (t CO2/yr), water retained (m3), temperature reduction (°C), species count. Economic: property value uplift (%), energy saved (kWh or CZK), avoided flood damage (CZK), ROI (%). Social: visitors per month, wellbeing score, number of community events, share of residents within 300 m (%).

4. What is each of these tools for: GIS, IoT sensors, i-Tree, a digital twin?

Recall

Model answer: GIS: maps with data layers (green cover, heat, access). Sensors: live field data (temperature, humidity, water flow). i-Tree: valuing ecosystem services of trees. Digital twin: a model of the place to test scenarios before investing.

5. What is Life Cycle Assessment and why is 'cradle to grave' important?

Understand

Model answer: LCA assesses environmental impacts for a defined function, scope and life cycle, using ISO 14040/14044 as the framework. It helps detect shifts between materials, production, use and end of life. The dominant stage depends on assumptions and the product.

6. Name the four phases of an LCA in order and say what happens in each.

Recall

Model answer: 1. Goal and scope: what is assessed, for which decision, with what boundaries. 2. Inventory analysis: collect all inputs and outputs. 3. Impact assessment: translate the inventory into impacts (climate, water, toxicity, land use). 4. Interpretation: conclusions, sensitivity checks, recommendations.

7. Explain the functional unit using the ceramic mug vs paper cup example.

Apply

Model answer: Use an equal service, such as serving a specified number and volume of drinks. Include manufacture, washing, reuse, breakage and disposal. The break-even point depends on these assumptions, so there is no universal number of uses.

8. Why is design the biggest lever for a product's environmental impact? Give two ecodesign principles for each of: materials, use phase, end of life.

Understand

Model answer: Design affects material quantities, manufacturing, energy use, repair and recovery. Examples include safer and separable materials; efficient, durable and repairable products; and recoverable components with take-back arrangements. A single fixed percentage should not be treated as universal.

9. Compare a green roof and a conventional roof over 50 years. Which burdens, benefits and assumptions could change the result?

Apply

Model answer: Compare equal waterproofing service over 50 years. Include structure, materials, replacement, maintenance, energy, water and disposal. A green roof may add load and initial costs while offering membrane protection, runoff retention, cooling and habitat. The preferred option depends on the site, assumptions and decision criteria; wider benefits should be reported separately where outside the LCA.

10. A company reports that its new product is 'greener' because the factory now uses solar power. What questions would a lifecycle thinker ask?

Apply

Model answer: Which stage dominates the product's impact - is it production at all? What happened to materials, transport, use and end of life? What is the functional unit? Has the impact simply moved to another stage (for example heavier materials that cost more energy in use)?

Handout version: questions only

1. Write the five steps of the measurement cycle in order and explain what a baseline is for.

2. Name the four qualities of a good KPI. Give one KPI that is easy to game and a better alternative.

3. Give three environmental, three economic and three social KPIs for a new park, each with a unit.

4. What is each of these tools for: GIS, IoT sensors, i-Tree, a digital twin?

5. What is Life Cycle Assessment and why is 'cradle to grave' important?

6. Name the four phases of an LCA in order and say what happens in each.

7. Explain the functional unit using the ceramic mug vs paper cup example.

8. Why is design the biggest lever for a product's environmental impact? Give two ecodesign principles for each of: materials, use phase, end of life.

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Source list for revised passages

Checked 5 October 2026. These sources support the revised passages; the original recommended reading and image credits remain in the deck. Company material is identified as company-reported evidence.

1. www.epa.gov - slides 20, 22, 23

https://www.epa.gov/heatislands/using-green-roofs-reduce-heat-islands

2. www.epa.gov - slides 23

https://www.epa.gov/green-infrastructure/economic-benefits-green-infrastructure